



PPE COMPLIANCE DETECTOR

Automated PPE Verification for Home Health Caregivers

ITAI 1378 — Midterm Project Proposal

TEAM MEMBERS

Huy Nguyen & Raymond Newton

PROJECT TIER

Tier 1

The Problem

- Home health caregivers are required to wear PPE (masks, gloves) during certain visits.
- Compliance today relies on self-reporting and occasional manual supervisor spot-checks.
- This does not scale across a large caregiver workforce and creates gaps during infection-control audits.

Who cares: Agency compliance officers, supervisors, and payer/UHC infection-control audits.

MANUAL CHECKS DON'T SCALE

1

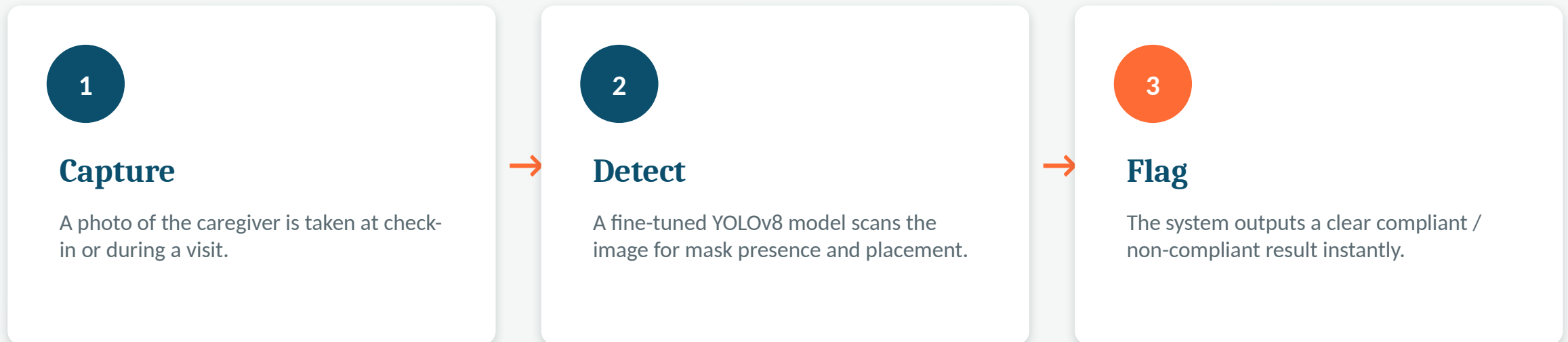
spot-check per visit — no continuous verification

Why it matters

Missed PPE compliance is both a health risk and a documentation liability during audits.

Our Solution

An object detection system that scans a photo of a caregiver and automatically flags PPE compliance — in real time, with no manual review.



Compliant → mask detected correctly **⚠ Non-Compliant** → mask missing or worn incorrectly

Technical Approach

TECHNIQUE	MODEL	FRAMEWORK
Object Detection Bounding boxes localize exactly where PPE is present or missing — not just a whole-image label.	YOLOv8n Fast to fine-tune on free GPU compute; YOLOv8s tested as a stretch comparison for accuracy vs. speed.	Ultralytics (PyTorch) Mature tooling, strong community support for PPE-style datasets, minimal boilerplate.

Why this stack: keeps scope realistic for the timeline — real training and evaluation, without custom architecture design or manual labeling.

Data Plan

DATASET SIZE

4,547

labeled images

Source

Roboflow Universe
“Mask-Detection-YOLOv8” (AGH)

License: CC BY 4.0

Labels (3 classes)

with_mask

without_mask

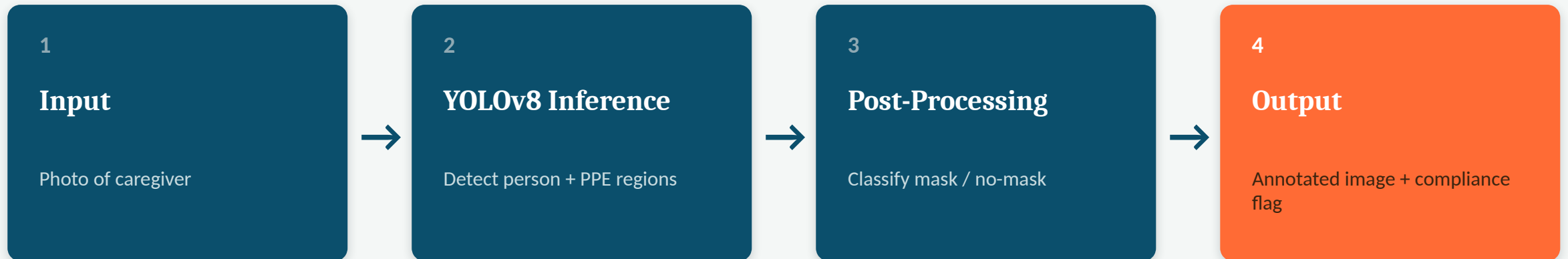
incorrectly_worn_mask

Core deliverable: with_mask / without_mask. Incorrect-mask detection is a stretch goal.

Preparation

- No manual labeling — dataset ships pre-annotated in YOLO format.
- Exports directly via the Roboflow Python package.
- Standard Ultralytics augmentation applied at train time (flip, mosaic, brightness).

System Diagram



Pipeline is intentionally linear — one photo in, one clear compliance flag out. No manual review step required.

Success Metrics

PRIMARY METRIC

mAP@0.5

$\geq 75-80\%$

Reference: a comparable model on this same dataset publicly reports ~91% mAP@50, supporting feasibility of this target.

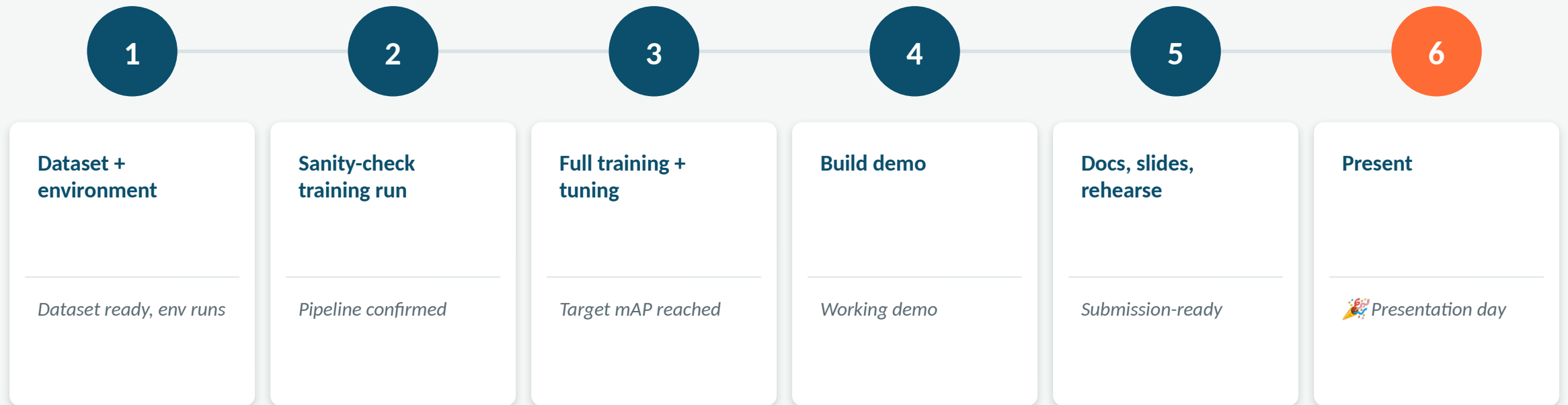
SECONDARY METRIC

Inference Speed

$< 0.5s$

per image, running on a free Colab T4 GPU — fast enough to demo near real-time.

Week-by-Week Plan



Risks & Mitigation

Risk	Probability	Mitigation
Class imbalance in dataset	Medium	Apply data augmentation; class-weighted loss if needed
Domain mismatch (dataset $\neq$ real visits)	Medium	Note as a known limitation; can't use real patient photos
Low mAP on full training run	Medium	More epochs; try YOLOv8s; add augmentation
Colab GPU unavailable / timeout	Medium	Checkpoint regularly; fall back to Kaggle Notebooks
Incomplete dataset annotations	Low	Dataset pre-validated on Roboflow; backup datasets available

Resources Needed

COMPUTE

Colab (free T4 GPU)

Kaggle Notebooks as backup

FRAMEWORKS

Ultralytics YOLOv8

PyTorch · Roboflow

ESTIMATED COST

\$0

Free tiers cover full project

A practical, scoped PPE compliance detector — built on public data, free compute, and a realistic 6-week timeline.

Thank you — questions welcome